Functional Requirements Document

PayTrack | Detailed behaviors, validations and controls

SAMPLE ONLY — FICTIONAL PORTFOLIO CASE STUDY. All names, records, targets, dates and amounts are illustrative, not evidence of actual client work.

Document version 1.0 | Scenario: PayTrack payment reconciliation portal | Owner: Sample Business Analyst

1. Purpose and relationship to BRD

This sample FRD translates BRD-PT-001 into observable system behaviors, business rules, data expectations and non-functional requirements. The functional design assumes a fictional internal operations portal and requires validation with a real product owner before implementation.

2. System context and actors

Actor / component	Primary interaction
Operations Analyst	Search transactions, open exceptions, add investigation notes and submit permitted resolutions.
Operations Lead	Review escalated cases, assign owners and approve controlled actions where configured.
Read-only Support	View permitted status and history.
Gateway Event Adapter	Receives / imports gateway events and normalizes source fields.
Reconciliation Service	Applies matching rules, writes reconciliation outcomes and creates exception cases.
Audit Service	Records actor, action, timestamp, target record and relevant change summary.

3. Functional requirements

ID	Area	Requirement / expected behavior	Priority	BRD trace
FR-001	Access control	The system shall require an authenticated user and enforce the assigned role permissions for each protected action.	Must	BR-05
FR-002	Overview metrics	The system shall show counts for received, matched, unmatched, pending and failed transactions for a selected time window.	Must	BR-01
FR-003	Transaction search	The system shall support search by payment ID, merchant reference and gateway reference.	Must	BR-06
FR-004	Transaction detail	The system shall show internal amount/currency/reference, gateway amount/currency/reference, current statuses and event timestamps when available.	Must	BR-01, BR-02
FR-005	Matching rule	The initial sample rule shall match on agreed reference plus amount and currency; exact key and tolerances require business approval.	Must	BR-02
FR-006	Reason codes	An unmatched record shall show a reason such as missing reference, amount mismatch, currency mismatch, duplicate event or timeout.	Must	BR-02

ID	Area	Requirement / expected behavior	Priority	BRD trace
FR-007	Exception creation	The system shall create or update one exception case per configured payment/reason combination and prevent duplicate open cases for the same key.	Must	BR-02
FR-008	Assignment	An authorised user shall assign a case to an eligible person or queue and the system shall record the change.	Must	BR-03, BR-04
FR-009	Investigation notes	An authorised user shall add a note or evidence reference without overwriting source transaction values.	Must	BR-03, BR-04
FR-010	Resolution	An authorised user shall select an allowed resolution type and provide a resolution reason before submitting.	Must	BR-03
FR-011	Approval / escalation	Configured high-risk or policy-defined resolutions shall require lead approval before status becomes resolved.	Should	BR-03, BR-05
FR-012	Audit trail	The system shall append an audit event for case creation, assignment, note, approval, resolution and status transition.	Must	BR-04
FR-013	Filtering	The system shall filter transaction and exception lists by status, date range, reason code and owner.	Should	BR-06
FR-014	Export	The system shall export currently filtered rows only to an approved format and omit fields the user is not authorised to view.	Could	BR-07
FR-015	Job status	The system shall expose ingestion and reconciliation job status, last-run time and a safe failure reason to authorised operations users.	Should	BR-08

4. Business rules and validations

Rule	Description
RULE-01	Amounts are compared using currency-aware decimal precision; no floating-point rounding rule is assumed.
RULE-02	A match is not inferred solely from amount. A stable agreed reference or composite key is required.
RULE-03	Duplicate events are idempotently ignored or linked to the existing transaction; behavior is configurable and must be confirmed.
RULE-04	Resolution submission requires resolution type, non-empty reason and the user permission to resolve.
RULE-05	Original gateway/internal values remain immutable in the evidence view; corrections are represented as new, auditable actions.
RULE-06	Status transitions must be valid: Open → In Review → Pending Approval (if required) → Resolved, or Escalated.
RULE-07	A resolved case may be reopened only by an authorised role and with a reason.
RULE-08	All timestamps are stored in a consistent standard and displayed in the configured user timezone.

5. Screen / interaction requirements

Screen	Required content	Primary actions
Reconciliation overview	Summary counts; trend visualization; exception categories; recent transactions; time-window filter.	Search, change time window, open transaction/case.
Transaction detail	Internal/gateway fields side by side; event chronology;	Open case, inspect event

Screen	Required content	Primary actions
	reconciliation outcome and reason.	references, copy identifiers.
Exception case detail	Case ID, status, owner, severity/reason, transaction comparison, notes, resolution form and audit history.	Assign, add note, resolve, escalate, request approval.
Exception queue	Case ID, payment ID, reason, age, owner, last update and status.	Filter, sort, assign, open case, export if permitted.

6. Data requirements

Entity	Sample key fields	Data notes
PaymentTransaction	paymentId, merchantReference, amount, currency, internalStatus, createdAt	Primary internal transaction record; source fields preserved.
GatewayEvent	eventId, paymentId, gatewayReference, eventType, receivedAt, payloadHash	One transaction may have multiple gateway events; sensitive raw payload access restricted.
ReconciliationResult	resultId, paymentId, matchStatus, mismatchCode, comparedAt, ruleVersion	Stores outcome of a comparison and the rule/version used.
ExceptionCase	caseId, paymentId, reasonCode, caseStatus, assignedTo, resolutionType, updatedAt	Open case deduplication depends on agreed uniqueness rule.
AuditEvent	auditId, targetType, targetId, actorId, action, timestamp, changeSummary	Append-only log; retention and export must follow approved policy.

7. Non-functional requirements (proposed)

ID	Requirement	Validation approach
NFR-01	Performance: key overview and queue views should render within 3 seconds at the 95th percentile under the agreed pilot load.	Performance test using agreed synthetic dataset and environment.
NFR-02	Availability: pilot target 99.5% monthly availability, excluding planned maintenance, pending platform confirmation.	Monitor availability and define outage calculation.
NFR-03	Security: least-privilege role access; secrets are not stored in the browser; audit-sensitive actions are server-authorized.	Role-based negative and positive tests; security review.
NFR-04	Reliability: repeated gateway events do not create duplicate financial actions or duplicate open cases.	Replay/idempotency tests.
NFR-05	Accessibility: core flows are keyboard reachable, labels are associated with fields, and status is not conveyed by color alone.	Keyboard walkthrough and automated/manual accessibility checks.
NFR-06	Privacy: use synthetic or masked information in non-production environments; define retention/export policy before live deployment.	Data review and security sign-off.

8. Error states and messages

- Source feed unavailable: show the latest successful run and a safe retry/escalation action.
- Required resolution reason missing: keep the case open and highlight the field; do not silently submit.

- Insufficient permission: deny server-side action and display a non-sensitive explanation.
- Duplicate or out-of-order event: retain an operationally useful reason/status without creating a duplicate financial record.
- Export failure: show a recoverable message and do not report a file as ready until generation succeeds.

9. Open decisions before build

Decision	Owner to confirm	Status
Authoritative matching key and amount tolerance	Product Owner + Finance/Ops	Open
Role-to-action matrix and approval thresholds	Operations Lead + Security	Open
Gateway source, event schema, retry strategy	Engineering	Open
Audit retention and permitted exports	Security / Compliance	Open
Pilot volume, load profile and performance target	Platform Owner	Open

10. Design references

Editable low-fidelity screens: https://www.figma.com/design/qt1PFmcQNvtaJnLnrWxPKD

Editable FigJam business process, API sequence and data model diagrams: https://www.figma.com/board/N90uAbHaqe4OkCOUkNfNxX